

Exact Results in Quantum Field Theory

Perimeter Scholars International, Quantum Field Theory II, Tutorial 4; 16 worked parts.

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Quantum Field Theory

Question prompts paraphrased from the official Perimeter PSI Online sheet; the equations and notation follow the source.

1. The Källén–Lehmann Representation

For a free scalar field, recall the Feynman propagator

$$D_0^F(x-y) \equiv \langle 0|T[\phi(x)\phi(y)]|0\rangle = -i \int \frac{d^4k}{(2\pi)^4} \frac{e^{ik \cdot (x-y)}}{k^2 + m_0^2}. \quad (1.1)$$

It describes propagation of a particle of mass m_0 . Its pole at $k^2 = -m_0^2$ is understood with the usual $-i\epsilon$ prescription. In an interacting theory, replace the free vacuum by the interacting vacuum and define

$$D^F(x-y) = \langle \Omega|T[\phi(x)\phi(y)]|\Omega\rangle. \quad (1.2)$$

The lowest pole of this propagator determines the physical mass. To obtain its nonperturbative spectral representation, insert a complete set of interacting states between the two fields.

- a) Assume Poincaré invariance, with four-momentum operator $P^\mu = (H, \mathbf{P})$ satisfying

$$[P^\mu, P^\nu] = 0. \quad (1.3)$$

Determine the action of translations on the field and the interacting vacuum:

$$e^{-iP \cdot x} \phi(0) e^{iP \cdot x} = ? \quad e^{-iP \cdot x} |\Omega\rangle = ? \quad (1.4)$$

Solution. The vacuum should be invariant under Poincaré transformation, namely $e^{-iP \cdot x} |\Omega\rangle = |\Omega\rangle$. It can also be shown that

$$e^{-iP \cdot x} \phi(0) e^{iP \cdot x} = \phi(x). \quad (1.5)$$

Take the states to be simultaneous energy–momentum eigenstates. Denote a one-particle state of physical mass m by $|\mathbf{k}\rangle$, with $E_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}$, and use $|\mathbf{k}, n\rangle$ for states containing two or more particles, where n collects the remaining quantum numbers.

- b) Write the spectral decomposition of the identity, including the vacuum, one-particle states, and multiparticle states.

Solution.

$$1 = |\Omega\rangle \langle \Omega| + \int \frac{d^3k}{(2\pi)^3 2E_{\mathbf{k}}} |\mathbf{k}\rangle \langle \mathbf{k}| + \sum_n \int \frac{d^3k}{(2\pi)^3 2E_{\mathbf{k}}} |\mathbf{k}, n\rangle \langle \mathbf{k}, n|. \quad (1.6)$$

- c) For $x^0 > y^0$, insert that identity on the right-hand side of the interacting propagator and show that

$$\langle \Omega| \phi(x) \phi(y) |\Omega\rangle \quad (1.7)$$

$$= \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{e^{ik \cdot (x-y)}}{2E_{\mathbf{k}}} |\langle \mathbf{k} | \phi(0) | \Omega \rangle|^2 + \sum_n \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{e^{ik \cdot (x-y)}}{2E_{\mathbf{k}}} |\langle \mathbf{k}, n | \phi(0) | \Omega \rangle|^2. \quad (1.8)$$

Assume that $\phi(x)|\Omega\rangle$ has a one-particle contribution.

Solution.

$$\begin{aligned}
\langle \Omega | \phi(x) \phi(y) | \Omega \rangle &= \langle \Omega | \phi(0) e^{iP \cdot (x-y)} \phi(0) | \Omega \rangle \\
&= \langle \Omega | \phi(0) e^{iP \cdot (x-y)} \left(|\Omega\rangle \langle \Omega| + \int \frac{d^3 \mathbf{k}}{(2\pi)^3 2E_{\mathbf{k}}} |\mathbf{k}\rangle \langle \mathbf{k}| \right. \\
&\quad \left. + \sum_n \int \frac{d^3 \mathbf{k}}{(2\pi)^3 2E_{\mathbf{k}}} |\mathbf{k}, n\rangle \langle \mathbf{k}, n| \right) \phi(0) | \Omega \rangle \\
&= \int \frac{d^3 \mathbf{k}}{(2\pi)^3} \frac{e^{ik \cdot (x-y)}}{2E_{\mathbf{k}}} |\langle \mathbf{k} | \phi(0) | \Omega \rangle|^2 + \sum_n \int \frac{d^3 \mathbf{k}}{(2\pi)^3} \frac{e^{ik \cdot (x-y)}}{2E_{\mathbf{k}}} |\langle \mathbf{k}, n | \phi(0) | \Omega \rangle|^2.
\end{aligned} \tag{1.9}$$

d) Introduce the spectral function

$$\rho(s) \equiv \delta(s - m^2) |\langle \mathbf{k} | \phi(0) | \Omega \rangle|^2 + \sum_n |\langle \mathbf{k}, n | \phi(0) | \Omega \rangle|^2 \delta(s - M_n^2), \quad M_n^2 = E_{\mathbf{k}}^2 - \mathbf{k}^2. \tag{1.10}$$

Assuming there are no bound states, show that $\rho(s) = 0$ for $s < 4m^2$ except at $s = m^2$, whereas $\rho(s)$ is nonzero for $s \geq 4m^2$. Use this result to establish, for $x^0 > y^0$,

$$\langle \Omega | \phi(x) \phi(y) | \Omega \rangle \tag{1.11}$$

$$\begin{aligned}
&= \int \frac{d^3 \mathbf{k}}{(2\pi)^3} \frac{e^{ik \cdot (x-y)}}{2E_{\mathbf{k}}} |\langle \mathbf{k} | \phi(0) | \Omega \rangle|^2 \\
&\quad + \int_{4m^2}^{\infty} ds \rho(s) \int \frac{d^3 \mathbf{k}}{(2\pi)^3} \frac{e^{ik \cdot (x-y)}}{2E_{\mathbf{k}}},
\end{aligned} \tag{1.12}$$

and, for $x^0 < y^0$,

$$\langle \Omega | \phi(y) \phi(x) | \Omega \rangle \tag{1.13}$$

$$\begin{aligned}
&= \int \frac{d^3 \mathbf{k}}{(2\pi)^3} \frac{e^{-ik \cdot (x-y)}}{2E_{\mathbf{k}}} |\langle \mathbf{k} | \phi(0) | \Omega \rangle|^2 \\
&\quad + \int_{4m^2}^{\infty} ds \rho(s) \int \frac{d^3 \mathbf{k}}{(2\pi)^3} \frac{e^{-ik \cdot (x-y)}}{2E_{\mathbf{k}}}.
\end{aligned} \tag{1.14}$$

Solution. If there are no bound states, then it is obvious that there are no additional poles for $s < 4m^2$. The result follows trivially.

e) Use the QFT I identity

$$\int \frac{d^4 k}{(2\pi)^4} \frac{e^{ik \cdot (x-y)}}{k^2 + m^2 - i\epsilon} \tag{1.15}$$

$$\begin{aligned}
&= i\Theta(x^0 - y^0) \int \frac{d^3 \mathbf{k}}{(2\pi)^3} \frac{e^{ik \cdot (x-y)}}{2E_{\mathbf{k}}} \\
&\quad + i\Theta(y^0 - x^0) \int \frac{d^3 \mathbf{k}}{(2\pi)^3} \frac{e^{-ik \cdot (x-y)}}{2E_{\mathbf{k}}}
\end{aligned} \tag{1.16}$$

to derive the Källén–Lehmann representation

$$\tilde{D}_F(k^2) = \frac{-i |\langle \mathbf{k} | \phi(0) | \Omega \rangle|^2}{k^2 + m^2 - i\epsilon} + \int_{4m^2}^{\infty} ds \rho(s) \frac{-i}{k^2 + s - i\epsilon}, \tag{1.17}$$

where $\tilde{D}_F$ is the Fourier transform of $D_F(x)$.

Solution. This follows trivially from repeating the same contour integral one used to obtain (1.16).

f) Regard $\tilde{D}_F(k^2)$ as an analytic function of k^2 . Locate its lowest pole and explain the physical meaning of both the pole and its residue.

Solution. The lowest pole is at $k^2 = -m^2$, which is the physical mass of the one-particle state, and the residue is the field renormalisation factor Z .

- g) The continuum begins at $s = 4m^2$ and produces a branch cut. If the theory also contains bound states, which determine where their contributions would appear in $\rho(s)$.

Solution. The bound states would still be single poles, but they would appear as δ -functions in the spectral function.

2. Schwinger–Dyson Equations

The Schwinger–Dyson equations state that correlation functions obey the classical equations of motion up to local contact terms.

- a) In the free Euclidean Klein–Gordon theory, use invariance of the path integral under $\phi(x) \mapsto \phi(x) + \epsilon(x)$. Starting from

$$\langle \phi(x) \rangle = \frac{1}{\mathcal{Z}[0]} \frac{\delta \mathcal{Z}[J]}{\delta J(x)} \Big|_{J=0}, \quad (2.1)$$

show that

$$(-\partial_z^2 + m^2) \langle \phi(z) \phi(x) \rangle = \hbar \delta(z - x). \quad (2.2)$$

Solution. Consider the integral under an infinitesimal field shifting $\phi \rightarrow \phi + \epsilon$

$$\int \mathcal{D}\phi F[\phi] e^{-S[\phi]/\hbar} = \int \mathcal{D}\phi F[\phi + \epsilon] e^{-S[\phi + \epsilon]/\hbar}. \quad (2.3)$$

Then, we have

$$\begin{aligned} 0 &= \int \mathcal{D}\phi \left[\left(F[\phi] + \frac{\delta F}{\delta \phi} \epsilon \right) \left(1 - \frac{1}{\hbar} \frac{\delta S}{\delta \phi} \epsilon \right) - F[\phi] \right] e^{-S[\phi]/\hbar} \\ &= \left\langle \frac{\delta F[\phi]}{\delta \phi} \right\rangle \epsilon - \frac{1}{\hbar} \left\langle F[\phi] \frac{\delta S[\phi]}{\delta \phi} \right\rangle \epsilon \end{aligned} \quad (2.4)$$

for all ϵ . Therefore, we find

$$\hbar \left\langle \frac{\delta F[\phi]}{\delta \phi} \right\rangle = \left\langle F[\phi] \frac{\delta S[\phi]}{\delta \phi} \right\rangle. \quad (2.5)$$

For our case, we take

$$S_0[\phi] = \int d^d z \frac{1}{2} \phi(z) (-\partial_z^2 + m^2) \phi(z), \quad F[\phi] = \phi(x), \quad (2.6)$$

which gives

$$(-\partial_z^2 + m^2) \langle \phi(z) \phi(x) \rangle = \hbar \delta(z - x). \quad (2.7)$$

- b) Add the interaction

$$S[\phi] = S_0[\phi] + \frac{\lambda}{n!} \int d^d y \phi^n(y), \quad (2.8)$$

and show that

$$(-\partial_z^2 + m^2) \langle \phi(z) \phi(x) \rangle = - \left\langle \frac{\lambda}{(n-1)!} \phi^{n-1}(z) \phi(x) \right\rangle + \hbar \delta(z - x). \quad (2.9)$$

Solution. Now, we have

$$\frac{\delta S[\phi]}{\delta \phi} = (-\partial_z^2 + m^2) \phi(z) + \frac{\lambda}{(n-1)!} \phi^{n-1}(z), \quad (2.10)$$

and hence, using the master equation, we find

$$(-\partial_z^2 + m^2)\langle\phi(z)\phi(x)\rangle = -\left\langle\frac{\lambda}{(n-1)!}\phi^{n-1}(z)\phi(x)\right\rangle + \hbar\delta(z-x). \quad (2.11)$$

- c) Use invariance of the generating functional under $\phi \mapsto \phi + \delta\phi$ to prove the general identity

$$\left\langle\frac{\delta S}{\delta\phi(x)}\prod_{r=1}^n\phi(x_r)\right\rangle = \hbar\sum_{j=1}^n\delta^4(x-x_j)\left\langle\prod_{\substack{r=1 \\ r\neq j}}^n\phi(x_r)\right\rangle. \quad (2.12)$$

What changes in real time?

Solution. This follows directly from substituting

$$F[\phi] = \prod_{r=1}^n\phi(x_r). \quad (2.13)$$

In real time, the expression becomes

$$\left\langle\frac{\delta S}{\delta\phi(x)}\prod_{r=1}^n\phi(x_r)\right\rangle = i\hbar\sum_{j=1}^n\delta^4(x-x_j)\left\langle\prod_{\substack{r=1 \\ r\neq j}}^n\phi(x_r)\right\rangle. \quad (2.14)$$

- d) Show that

$$\langle\phi_1\cdots\phi_n\rangle = \int d^4x\Delta_{x1}(-\partial_x^2 + m^2)\langle\phi_x\phi_2\cdots\phi_n\rangle, \quad (2.15)$$

where $\phi_i = \phi(x_i)$ and $\Delta_{ij} = \Delta_F(x_i - x_j)$.

Solution. Using $(-\partial_x^2 + m^2)\Delta_{1x} = \delta_{1x}$, we obtain the above equation.

- e) In the free Euclidean Klein–Gordon theory, combine the preceding result with the Schwinger–Dyson equations to obtain

$$\langle\phi_1\cdots\phi_4\rangle_0 = \Delta_{12}\Delta_{34} + \Delta_{13}\Delta_{24} + \Delta_{14}\Delta_{23}. \quad (2.16)$$

Solution.

$$\begin{aligned} \langle\phi_1\phi_2\phi_3\phi_4\rangle_0 &= \int d^d x\Delta_{x1}(-\partial_x^2 + m^2)\langle\phi_x\phi_2\phi_3\phi_4\rangle_0 \\ &= \int d^d x\Delta_{x1}(\delta_{x2}\Delta_{34} + \delta_{x3}\Delta_{24} + \delta_{x4}\Delta_{23}) \\ &= \Delta_{12}\Delta_{34} + \Delta_{13}\Delta_{24} + \Delta_{14}\Delta_{23}, \end{aligned} \quad (2.17)$$

where we will take $\hbar = 1$ from now on.

- f) In Euclidean ϕ^3 theory, apply the same identities recursively and show that

$$\langle\phi_1\phi_2\rangle \quad (2.18)$$

$$= \Delta_{12} + (-g)^2 \int d^4x d^4y \left(\frac{1}{2}\Delta_{x1}\Delta_{xy}^2\Delta_{y2} + \frac{1}{4}\Delta_{x1}\Delta_{xx}\Delta_{yy}\Delta_{y2} \right) + O(g^4). \quad (2.19)$$

Does this agree with the result obtained in Tutorial 2?

Solution.

$$\begin{aligned}
\langle \phi_1 \phi_2 \rangle &= \int d^d x \Delta_{1x} (-\partial_x^2 + m^2) \langle \phi_x \phi_2 \rangle \\
&= \int d^d x \Delta_{1x} \left(-\frac{g}{2} \langle \phi_x^2 \phi_2 \rangle + \delta_{x2} \right) \\
&= \Delta_{12} - \frac{g}{2} \int d^d x d^d y \Delta_{1x} \Delta_{2y} (-\partial_y^2 + m^2) \langle \phi_x^2 \phi_y \rangle \\
&= \Delta_{12} - \frac{g}{2} \int d^d x d^d y \Delta_{1x} \Delta_{2y} \left(-\frac{g}{2} \langle \phi_y^2 \phi_x^2 \rangle + 2\delta_{xy} \langle \phi_x \rangle \right).
\end{aligned} \tag{2.20}$$

Since we are only interested to order g^2 , we have

$$\langle \phi_y^2 \phi_x^2 \rangle = \langle \phi_y^2 \phi_x^2 \rangle_0 + \mathcal{O}(g^2) = 2\Delta_{xy}^2 + \Delta_{xx} \Delta_{yy} + \mathcal{O}(g^2). \tag{2.21}$$

Meanwhile, the tadpole diagram is given by

$$\begin{aligned}
\langle \phi_x \rangle &= \int d^d z \Delta_{xz} (-\partial_z^2 + m^2) \langle \phi_z \rangle \\
&= -\frac{g}{2} \int d^d z \Delta_{xz} \langle \phi_z^2 \rangle \\
&= -\frac{g}{2} \int d^d z \Delta_{xz} \Delta_{zz} + \mathcal{O}(g^3).
\end{aligned} \tag{2.22}$$

Then, we have

$$\begin{aligned}
\langle \phi_1 \phi_2 \rangle &= \Delta_{12} + (-g)^2 \int d^d x d^d y \Delta_{1x} \Delta_{2y} \left(\frac{1}{2} \Delta_{xy}^2 + \frac{1}{4} \Delta_{xx} \Delta_{yy} \right) \\
&\quad + (-g)^2 \int d^d x d^d y \frac{1}{2} \Delta_{1x} \Delta_{2x} \Delta_{xy} \Delta_{yy} + \mathcal{O}(g^4).
\end{aligned} \tag{2.23}$$

g) **Optional.** For a continuous symmetry of the Lagrangian, prove the Ward–Takahashi identity

$$\begin{aligned}
\partial_\mu \langle j^\mu(x) \phi(x_1) \cdots \phi(x_n) \rangle &= \hbar \sum_{j=1}^n \langle \phi(x_1) \cdots \phi(x_{j-1}) \\
&\quad \times \delta \phi(x) \delta^4(x - x_j) \phi(x_{j+1}) \cdots \phi(x_n) \rangle,
\end{aligned} \tag{2.24}$$

where, in the notation of the sheet,

$$j^\mu(x) = \frac{\delta S}{\delta \phi_{,\mu}(x)} \delta \phi(x) - \delta \mathcal{L}(x), \quad \phi_{,\mu} \equiv \partial_\mu \phi. \tag{2.25}$$

What changes in real time?

Solution. The classical action has some global symmetry, $\phi(x) \rightarrow \phi(x) + \epsilon(x) \delta \phi(x)$, which corresponds to a Noether current, $j_\mu(x)$. Now, we promote the symmetry to be a local symmetry $\phi(x) \rightarrow \phi(x) + \epsilon(x) \delta \phi(x)$. Again, we consider the path integral

$$\int \mathcal{D}\phi F[\phi] e^{-S[\phi]/\hbar} = \int \mathcal{D}\phi (F[\phi] + \delta F[\phi]) \left(1 - \frac{1}{\hbar} \delta S[\phi] \right) e^{-S[\phi]/\hbar}, \tag{2.26}$$

where $F[\phi] \rightarrow F[\phi] + \delta F[\phi]$ under the symmetry transformation. Note that $\delta S, \delta F$ are of order ϵ , and for an infinitesimal transformation, we can ignore terms of order ϵ^2 . Then, we have

$$\hbar \langle \delta F[\phi] \rangle = \langle \delta S[\phi] F[\phi] \rangle. \tag{2.27}$$

Recall

$$\begin{aligned}
\delta S &= \int d^d x \left[\frac{\partial \mathcal{L}}{\partial \phi(x)} \epsilon(x) \delta \phi(x) + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi(x))} \partial_\mu (\epsilon(x) \delta \phi(x)) \right] \\
&= \int d^d x \epsilon(x) \left[\frac{\partial \mathcal{L}}{\partial \phi(x)} \delta \phi(x) + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi(x))} \partial_\mu \delta \phi(x) \right] + \int d^d x \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi(x))} \delta \phi(x) \partial_\mu \epsilon(x) \\
&= \int d^d x \left[\epsilon(x) \partial_\mu \delta \mathcal{L}^\mu(x) + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi(x))} \delta \phi(x) \partial_\mu \epsilon(x) \right] \\
&= \int d^d x \partial_\mu (\epsilon(x) \delta \mathcal{L}^\mu(x)) + \int d^d x \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi(x))} \delta \phi(x) - \delta \mathcal{L}^\mu(x) \right] \partial_\mu \epsilon(x) \\
&= \int d^d x j^\mu(x) \partial_\mu \epsilon(x) \\
&= - \int d^d x \epsilon(x) \partial_\mu j^\mu(x).
\end{aligned} \tag{2.28}$$

and take

$$\begin{aligned}
F[\phi] = \phi(x_1) \cdots \phi(x_n) &\implies \delta F[\phi] = \sum_{j=1}^n \phi(x_1) \cdots \phi(x_{j-1}) \epsilon(x_j) \delta \phi(x_j) \phi(x_{j+1}) \cdots \phi(x_n) \\
&= \sum_{j=1}^n \int d^d x \delta^d(x - x_j) \phi(x_1) \cdots \phi(x_{j-1}) \epsilon(x) \delta \phi(x) \phi(x_{j+1}) \cdots \phi(x_n).
\end{aligned} \tag{2.29}$$

Hence, we find the Ward-Takahashi identity

$$\partial_\mu \langle j^\mu(x) \phi(x_1) \cdots \phi(x_n) \rangle = - \hbar \sum_{j=1}^n \langle \phi(x_1) \cdots \phi(x_{j-1}) \delta \phi(x) \delta^4(x - x_j) \phi(x_{j+1}) \cdots \phi(x_n) \rangle. \tag{2.30}$$

In real time, this becomes

$$\partial_\mu \langle j^\mu(x) \phi(x_1) \cdots \phi(x_n) \rangle = - i \hbar \sum_{j=1}^n \langle \phi(x_1) \cdots \phi(x_{j-1}) \delta \phi(x) \delta^4(x - x_j) \phi(x_{j+1}) \cdots \phi(x_n) \rangle. \tag{2.31}$$

h) **More optional.** Show that contact terms on the right-hand side of the Schwinger–Dyson and Ward–Takahashi identities do not contribute to $i\mathcal{M}$. In Euclidean momentum conventions, use

$$\langle f|S|i \rangle = \lim_{k_i^2 \rightarrow -m^2} (k_i^2 + m^2) \cdots \langle \Omega | T \tilde{\phi}(k_1) \cdots | \Omega \rangle, \tag{2.32}$$

with

$$\tilde{\phi}(k) = i \int d^4 x e^{-ik \cdot x} \phi(x). \tag{2.33}$$

The scattering amplitude and momentum-space correlator contain the same momentum-conserving delta function:

$$\langle f|S|i \rangle = (2\pi)^4 \delta^4\left(\sum k\right) i\mathcal{M}, \tag{2.34}$$

$$\langle \Omega | T \tilde{\phi}(k_1) \cdots | \Omega \rangle = (2\pi)^4 \delta^4\left(\sum k\right) \mathcal{F}(k_i). \tag{2.35}$$

Near the simultaneous poles,

$$\mathcal{F}(k_i) = \frac{i\mathcal{M}}{(k_1^2 + m^2) \cdots (k_n^2 + m^2)} + \text{nonsingular terms}. \tag{2.36}$$

Argue that the Fourier transform of a contact term lacks the required multivariable pole and hence cannot contribute to $i\mathcal{M}$.

Solution. Consider a contact term

$$\delta^d(x_1 - x_2) \langle \phi(x_3) \cdots \phi(x_n) \rangle, \quad (2.37)$$

which is Fourier transformed to

$$(2\pi)^d \delta^d(k_1 + k_2) \langle \tilde{\phi}(k_3) \cdots \tilde{\phi}(k_n) \rangle. \quad (2.38)$$

Its contribution to the LSZ is

$$\lim_{k_i^2 \rightarrow -m^2} (k_1^2 + m^2) \cdots (k_n^2 + m^2) (2\pi)^d \delta^d(k_1 + k_2) \langle \tilde{\phi}(k_3) \cdots \tilde{\phi}(k_n) \rangle. \quad (2.39)$$

Since it contains no poles in k_1, k_2 , it has no contribution to the LSZ reduction formula.

i) **More optional.** In Lorenz gauge, use the photon LSZ formula

$$\langle f|S|i \rangle = \epsilon^\mu \int d^4x e^{-ik \cdot x} (-\partial^2) \cdots \langle \Omega | T A_\mu(x) \cdots | \Omega \rangle \quad (2.40)$$

together with the Ward-Takahashi identity to derive

$$k_\mu \mathcal{M}^\mu = 0, \quad \mathcal{M} = \epsilon_\mu \mathcal{M}^\mu. \quad (2.41)$$

Solution. In QED, we have the equation of motion for the gauge fields in the Lorenz gauge

$$-\partial^2 A_\mu = j_\mu = e \bar{\psi} \gamma_\mu \psi. \quad (2.42)$$

The current corresponds to the global $U(1)$ transformation. The LSZ becomes

$$\langle f|S|i \rangle = \epsilon^\mu \int d^4x e^{-ik \cdot x} \langle \Omega | T j_\mu(x) \cdots | \Omega \rangle = (2\pi)^d \delta^d \left(\sum k \right) \epsilon_\mu \mathcal{M}_\mu. \quad (2.43)$$

Now replacing $\epsilon^\mu \rightarrow k^\mu$, we have

$$\begin{aligned} (2\pi)^d \delta^d \left(\sum k \right) k_\mu \mathcal{M}_\mu &= i \int d^4x (\partial^\mu e^{-ik \cdot x}) \langle \Omega | T j_\mu(x) \cdots | \Omega \rangle \\ &= -i \int d^4x e^{-ik \cdot x} \partial^\mu \langle \Omega | T j_\mu(x) \cdots | \Omega \rangle. \end{aligned} \quad (2.44)$$

Now, since we have shown that the contact terms do not contribute to S -matrix, and the LHS of the Ward-Takahashi identity is purely contact terms. Therefore, we can conclude that $k_\mu \mathcal{M}^\mu = 0$.